INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Meta Henrico Data Center, VA -- station KRIC, America/New_York
Committed pair	OSM 1493516628 -> 1493516625 RIC2 DC6 / RIC2 DC3
Facade gap	195.8 m between the two halls
Weather record	43,736 real hourly records from KRIC
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3324 C at 215 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-03 (safe_sector)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 11 of 24 hours with 2 mode changes. 10 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	11 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 10 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	9.055	18.0	9.440	4.151
01	FREE	yes	-	9.068	18.0	8.330	3.272
02	FREE	yes	-	9.583	18.0	7.220	3.003
03	FREE	yes	-	9.597	18.0	6.670	2.704
04	FREE	yes	-	9.754	18.0	6.670	2.566
05	FREE	yes	-	10.180	18.0	6.110	2.745
06	FREE	yes	-	10.177	18.0	6.110	2.235
07	FREE	yes	-	10.821	18.0	6.110	2.555

08	FREE	yes	-	12.432	18.0	6.110	3.066
09	FREE	yes	-	14.664	18.0	6.110	3.747
10	FREE	yes	-	17.350	18.0	6.670	4.689
11	mechanical	no	dry-bulb	18.508	18.0	6.670	5.066
12	mechanical	no	dry-bulb	19.569	18.0	6.670	5.360
13	mechanical	no	dry-bulb	19.169	18.0	6.670	5.434
14	mechanical	yes	switch budget	16.499	18.0	6.670	4.302
15	mechanical	yes	switch budget	14.362	18.0	6.670	3.888
16	mechanical	yes	switch budget	12.351	18.0	6.110	3.555
17	mechanical	yes	switch budget	10.213	18.0	6.110	3.569
18	mechanical	yes	switch budget	7.909	18.0	6.110	3.297
19	mechanical	yes	switch budget	6.332	18.0	6.110	3.580
20	mechanical	yes	switch budget	5.139	18.0	6.110	3.761
21	mechanical	yes	switch budget	5.177	18.0	5.000	4.393
22	mechanical	yes	switch budget	4.664	18.0	5.000	4.404
23	mechanical	yes	switch budget	5.213	18.0	4.440	4.405

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.055 C, which is 8.945 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.151 C, and the margin is measured, not chosen: 3.976 C of group-conditional forecast error for this hour of day, plus 0.1748 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.068 C, which is 8.932 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.272 C, and the margin is measured, not chosen: 3.094 C of group-conditional forecast error for this hour of day, plus 0.1779 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.583 C, which is 8.417 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.003 C, and the margin is measured, not chosen: 2.860 C of group-conditional forecast error for this hour of day, plus 0.1432 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.597 C, which is 8.403 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.704 C, and the margin is measured, not chosen: 2.547 C of group-conditional forecast error for this hour of day, plus 0.1566 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.753 C, which is 8.247 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.566 C, and the margin is measured, not chosen: 2.261 C of group-conditional forecast error for this hour of day, plus 0.3047 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.180 C, which is 7.820 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.745 C, and the margin is measured, not chosen: 2.555 C of group-conditional forecast error for this hour of day, plus 0.1902 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.177 C, which is 7.823 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.235 C, and the margin is measured, not chosen: 2.057 C of group-conditional forecast error for this hour of day, plus 0.1772 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.821 C, which is 7.179 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.555 C, and the margin is measured, not chosen: 2.284 C of group-conditional forecast error for this hour of day, plus 0.2709 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.433 C, which is 5.567 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.065 C, and the margin is measured, not chosen: 2.853 C of group-conditional forecast error for this hour of day, plus 0.2125 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.664 C, which is 3.336 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.747 C, and the margin is measured, not chosen: 3.533 C of group-conditional forecast error for this hour of day, plus 0.2140 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.350 C, which is 0.650 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.689 C, and the margin is measured, not chosen: 4.559 C of group-conditional forecast error for this hour of day, plus 0.1302 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.508 C against a 18.0 C limit, so it fails by 0.508 C. It would take a limit 0.508 C higher, or a bound 0.508 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.568 C against a 18.0 C limit, so it fails by 1.568 C. It would take a limit 1.568 C higher, or a bound 1.568 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.169 C against a 18.0 C limit, so it fails by 1.169 C. It would take a limit 1.169 C higher, or a bound 1.169 C tighter, to change this hour.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.499 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.362 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.351 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.213 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.908 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.333 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.139 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.177 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.664 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.213 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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